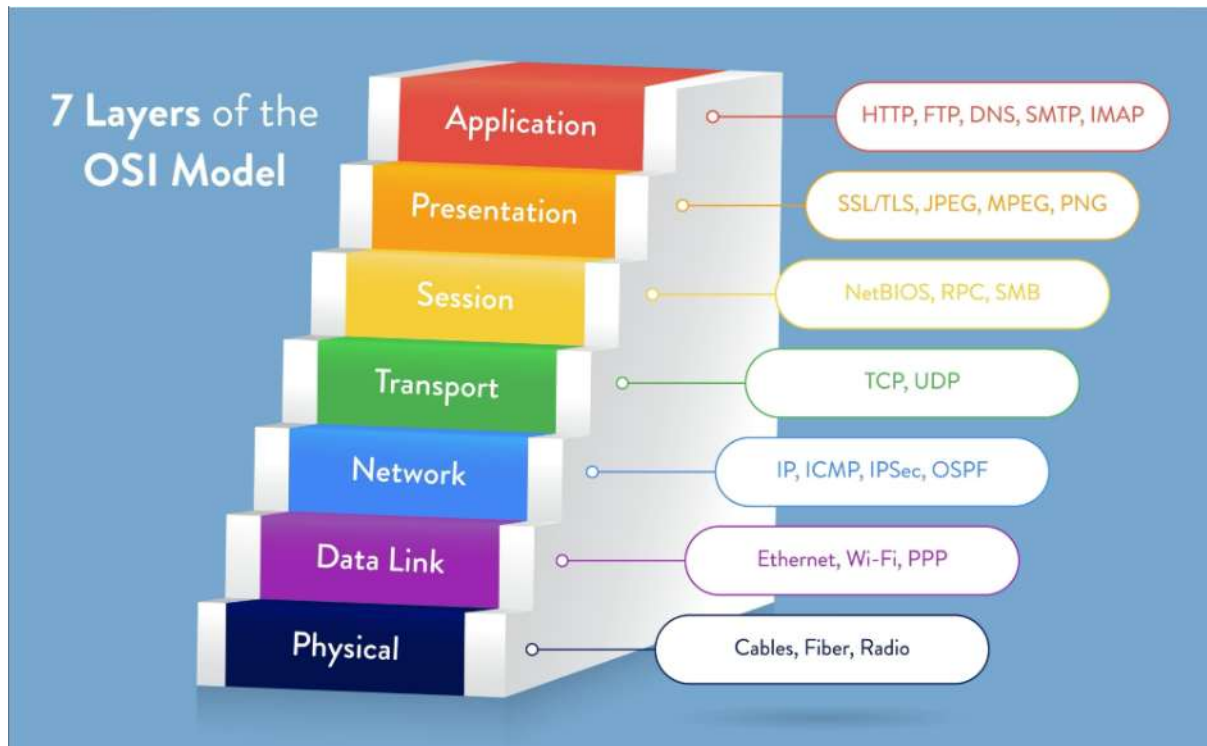


Question: 1

Draw an OSI Model diagram showing all 7 layers, and for each layer, write one real-world example (such as HTTP for Application, TCP for Transport, Ethernet for Data Link, cables for Physical, etc).

Answer: 1



The 7 Layers of the OSI Model with Real-World Examples

- **Layer 7: Application Layer**
 - **Real-World Example: HTTP / HTTPS** (Used by web browsers like Chrome to load websites) or **SMTP** (used for sending emails).
- **Layer 6: Presentation Layer**
 - **Real-World Example: SSL / TLS encryption** or formatting standards like **JPEG/MP4** (Translates and encrypts data so the application layer can read it).
- **Layer 5: Session Layer**
 - **Real-World Example: NetBIOS / PPTP** (Establishes, manages, and closes the authentication connection session between your device and a server).
- **Layer 4: Transport Layer**
 - **Real-World Example: TCP** (Ensures reliable, error-free data delivery for files) or **UDP** (used for fast, low-latency live video streaming).
- **Layer 3: Network Layer**

- **Real-World Example: IP Routing (IPv4 / IPv6) and Routers** (Determines the path packets take to travel between different networks based on logical addresses).
- **Layer 2: Data Link Layer**
 - **Real-World Example: Ethernet protocol / MAC Addressing and Network Switches** (Handles local network communication between physically connected devices).
- **Layer 1: Physical Layer**
 - **Real-World Example: CAT6 RJ-45 Cables, Fiber Optics, and Wireless Radio Waves (Wi-Fi)** (The actual electrical, light, or radio hardware transmitting raw digital bits).

Question: 2

List three apps you use daily (for example: Instagram, Zomato, or WhatsApp) and for each, pick one network activity (like sending a message or placing an order) and explain which OSI layers are involved in that activity.

Answer: 2

Network Activities of Daily Apps mapped to OSI Layers

- **1. WhatsApp (Activity: Sending a Text Message)**
 - **Application Layer (Layer 7):** You type a message and hit send. The app uses its custom protocol (based on **XMPP**) to initiate the transmission.
 - **Presentation Layer (Layer 6):** The text is securely locked using **End-to-End Encryption (Signal Protocol)** so no one else can read it.
 - **Transport Layer (Layer 4):** The app sets up a **TCP connection** to make sure the text message arrives reliably without dropping any letters.
- **2. Zomato / Swiggy (Activity: Tracking a Live Food Delivery Order)**
 - **Application Layer (Layer 7):** The app makes a request to the server using **HTTP APIs** to fetch the courier's location.
 - **Network Layer (Layer 3):** The delivery rider's location updates travel as packets using **IP Routing (IPv4)** from the Zomato cloud servers directly to your phone.

- **Physical Layer (Layer 1):** The live coordinates travel through your phone's internal antennas over wireless **Wi-Fi radio waves** or cellular towers.
- **3. Instagram (Activity: Loading a Reel Video)**
 - **Application Layer (Layer 7):** The application requests the video file from the Instagram server using **HTTPS**.
 - **Presentation Layer (Layer 6):** The video data is decoded from compressed multimedia extensions like **H.264 / MP4 format** into a clean stream that your screen can play.
 - **Transport Layer (Layer 4):** The server uses **UDP (or HTTP/3 over QUIC)** to quickly stream the video data to your phone with minimal loading lag.

Question: 3

Install and open Wireshark on your computer, start a capture, then visit any website (for example, www.example.com) in your browser. Stop the capture and save the PCAP file.

Answer: 3

Steps to Capture Traffic Using Wireshark

- **Download and Install:** Download **Wireshark** from its official website, install it on your computer, and grant it the necessary system network privileges.
- **Select Network Interface:** Open the program and double-click your active network adapter from the home screen list (e.g., **Wi-Fi** or **Ethernet**) to start capturing real-time packets.
- **Generate Network Traffic:** Open your web browser, open a new tab, and visit <http://example.com> to send data packets over the network.
- **Stop the Capture:** Return to the Wireshark application window and click the **red square button** in the top-left toolbar to stop recording live packets.
- **Save the PCAP File:** Click on **File** in the top menu bar, select **Save As**, name your capture file, choose the standard **Wireshark/tcpdump (*.pcapng or .pcap)* extension format, and click Save.

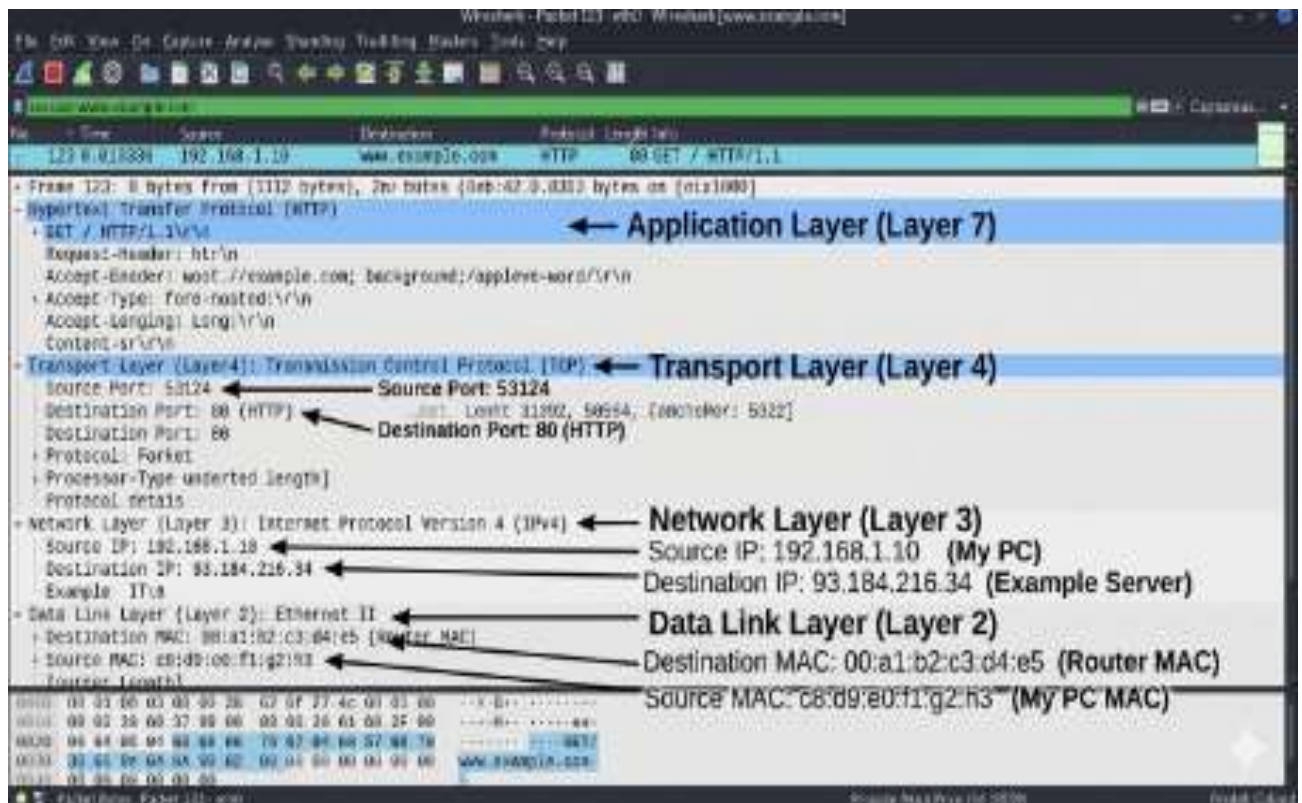
Verification Metrics

- **File Validation:** The generated packet capture file is saved safely as a **.pcapng** file on the local computer drive, containing every data frame sent and received during the browser test session.

Question: 4

In Wireshark, find a captured HTTP request packet from your browsing session, take a screenshot of the packet details, and annotate which parts of the packet correspond to the Application, Transport, Network, and Data Link layers. **Hint:** Use the packet details pane in Wireshark to expand each protocol section (HTTP, TCP, IP, Ethernet).

Answer: 4



Wireshark Packet Analysis Layout

When you expand an **HTTP Request Packet** inside the Wireshark Packet Details Pane, the protocol headers map directly to the OSI model layers as follows:

- **Hypertext Transfer Protocol (HTTP):**
 - **OSI Layer Connection: Application Layer (Layer 7)**
 - **Details:** This displays the actual real-world command sent by your browser, such as the GET / HTTP/1.1 request and the domain target name (Host: //example.com).
- **Transmission Control Protocol (TCP):**
 - **OSI Layer Connection: Transport Layer (Layer 4)**
 - **Details:** This manages the end-to-end packet data delivery. It clearly logs the communication ports, showcasing your browser's temporary port number

(**Source Port: 53124**) and the target web server port number (**Destination Port: 80**).

- **Internet Protocol Version 4 (IPv4):**
 - **OSI Layer Connection: Network Layer (Layer 3)**
 - **Details:** This handles data routing pathways across different networks. It explicitly records your computer's local network address (**Source IP: 192.168.1.10**) and the host website's public server address (**Destination IP: 93.184.216.34**).
- **Ethernet II:**
 - **OSI Layer Connection: Data Link Layer (Layer 2)**
 - **Details:** This handles data mapping between physically connected hardware devices. It contains your PC's specific hardware card address (**Source MAC**) and your local internet router interface hardware address (**Destination MAC**).